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What drives the TIPS–Treasury bond mispricing?

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ABSTRACT

Inflation-swapped Treasury Inflation-Protected Securities (TIPS) are usually undervalued compared to cash flow-matched Treasury bonds. From 2005 to 2022, TIPS discounts are persistent, averaging approximately 3.18 % of the face value, with a peak of 16.10 %. We elucidate the factors associated with this persistent mispricing and the extent of this association. The results from feature selection techniques and the variable importance-in-projection method reveal that marking-to-market concerns and intermediation frictions are key to understanding the underpricing of TIPS relative to comparable nominal Treasury securities. We conclude that when strategic concerns overwhelm fundamental analysis, asset prices could deviate from fundamental values over a prolonged period.

1. Introduction

Aside from nominal Treasury securities, the U.S. government furnishes other types of fixed-income instruments, such as Treasury inflation-protected securities (TIPS) and Treasury floating-rate notes. TIPS allow investors to infer inflation expectations because they are designed to protect investors against realized inflation, as the face value and coupon payments are linked to the consumer price index (CPI).[1] Hence, the difference between the nominal Treasury securities yield and the real yield of TIPS can be a measure of market-based inflation expectations known as break-even inflation (BEI). Moreover, inflation swaps that aim to hedge or speculate on inflation risks are widely traded in over-the-counter (OTC) derivatives markets, thereby offering more direct vehicles for trading or hedging inflation expectations.[2] In practice, inflation swap rates are almost always substantially higher than BEI (Fleming and Sporn, 2013)—this is dubbed as the TIPS–Treasury bond puzzle (i.e., the underpricing of TIPS relative to Treasury bonds). This mispricing in TIPS compared to cash-flow-matched and inflation-swapped Treasury bonds violates the law of one price (Fleckenstein et al., 2014).

TIPS discounts in the early years of the TIPS program can be attributed to liquidity premium (Haubrich et al., 2012; d'Amico et al., 2018; Driessen et al., 2017; Andreasen et al., 2021; Christensen and Gillan, 2012).[3] The price discounts owing to market illiquidity from the late 1990s to the early 2000s are considered "growing pains." The violation of the no-arbitrage principle in the TIPS market could be owing to slow-moving capital (Mitchell et al., 2007; Duffie, 2010; Mitchell and Pulvino, 2012; Fleckenstein et al., 2014).

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[1] The Bureau of Labor Statistics of the U.S. Department of Labor publishes the non-seasonally adjusted U.S. city average of all items for all urban consumers.

[2] In inflation swaps, payers receive fixed swap strike rates in exchange for realized inflation rates.

[3] The first TIPS auction was held on January 29, 1997.

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However, since TIPS remain considerably underpriced compared to their nominal counterparts, we posit that TIPS discounts could be driven by factors other than liquidity or slow-moving capital.

This study empirically investigates the potential channels associated with the TIPS–Treasury bond puzzle. We delineate the forces that could affect the persistent and significant underpricing of TIPS issues relative to nominal Treasury instruments. We consider various candidate metrics that influence price fluctuations in the fixed-income market, including Treasury inconvenience yields (He et al., 2022), ex-ante volatility measures (Britten-Jones and Neuberger, 2000; Bakshi and Madan, 2000; Jiang and Tian, 2005; Carr and Wu, 2009), funding liquidity and margin constraints (Brunnermeier and Pedersen, 2009; Garleanu and Pedersen, 2011), and repo specialness (Duffie, 1996; Jordan and Jordan, 1997; Krishnamurthy, 2002; Banerjee and Graveline, 2013). We utilize J.P. Morgan's proprietary asset swaps data from 2005 to 2022 on a daily frequency and introduce a simple and readily tradable way to quantify the magnitude of TIPS discounts over Treasury bonds. We find that all our empirical proxies for the TIPS–Treasury bond puzzle are significantly and positively associated with TIPS discounts (p-value < 0.01). As the potential drivers are highly correlated in the data, we use principal component analysis to obtain a low-dimensional and linearly independent representation of the explanatory variables space. We extract two principal components that capture the variations in the correlation matrix of the potential drivers and name the two factors "marking-to-market concerns" (Liu and Longstaff 2004, among others) and "intermediation frictions" (He and Krishnamurthy 2013; Adrian and Shin 2014; He et al., 2017), respectively, following their economic interpretations. To identify the key driving forces of TIPS discounts, we employ the adaptive elastic net (Zou and Hastie, 2005). As a robustness check, we further implement the variable-importance-in-projection method (Wold, 1966, 1975, 1994) and corroborate the findings of the feature selection method. In addition, we conduct Bai and Perron's (1998, 2003) test for structural breaks and document the time-varying relationship between potential drivers and TIPS discounts for each regime. The empirical evidence suggests that TIPS discounts are mainly associated with marking-to-market concerns; although, the impact of intermediation frictions in the fixed-income market regarding TIPS discounts is also substantial.

This study shows that when strategic concerns overwhelm fundamental analysis, asset prices could deviate from equilibrium values for a prolonged period. The literature shows that arbitrage opportunities could exist for a prolonged period because of a) fundamental risks that arbitragers face when implementing arbitrage transactions (Shleifer and Vishny 1997), b) various costs and market frictions (Gromb and Vayanos, 2010), c) slow-moving capital (Mitchell et al., 2007; Duffie, 2010; Mitchell and Pulvino, 2012), or d) portfolio constraints that prohibit textbook arbitrage opportunities from being exploited (Liu and Longstaff, 2004).[4] We provide a practical example of hurdles that impede textbook arbitrage transactions.

The remainder of this paper is organized as follows. Section 2 discusses issues regarding the pricing of the TIPS. Section 3 details our framework for quantifying the magnitude of the TIPS–Treasury bond puzzle and introduces potential drivers that could influence the mispricing. Section 4 presents the data and results. Section 5 concludes.

2. The pricing of TIPS

TIPS are designed to protect investors against inflation in the long term. In this section, we discuss inflation issues surrounding TIPS and summarize the illiquidity issues in the early stages of the TIPS program. We then revisit the TIPS–Treasury bond puzzle and introduce potential channels that could be associated with the puzzle.

2.1. TIPS and inflation risk premium

Because the principal amount and coupon payments of a TIPS issue adjust in proportion to realized inflation, TIPS bear no risk of inflation. Contrastingly, nominal Treasury yields contain inflation risk premium. The advantage of TIPS having no inflation risk premium could lower the issuer's expected borrowing cost, serving as a main reason for the launching of inflation-indexed bonds in the late 1990s.

The literature on inflation risk premium widely documents the presence of compensation for the risk of inflation. The inflation risk premium appears stable over time (Adrian and Wu, 2009) and amounts to 50–100 (20–140) basis points for the 5-year (10-year) horizon (Campbell and Shiller, 1996; Buraschi and Jiltsov, 2005). Grishchenko and Huang (2013) document that the inflation risk premium inferred from TIPS prices is time-varying over the period 2000–2008. Ang et al. (2008) report that variations in expected inflation and inflation risk premium account for approximately 80 % of the time-varying inflation compensation. Put differently, the prices of nominal Treasury securities are deflated owing to the inflation risk premium because nominal Treasury yields include the risk compensation for fluctuations in expected inflation by such degrees. In turn, the U.S. Treasury Department should bear the deflated prices that emanate from positive and substantial inflation risk premium. Other things being equal, issuers of nominal Treasury instruments benefit from eliminating inflation risk premium, thereby increasing Treasury securities prices and lowering their borrowing costs (Fleming and Krishnan, 2012). This caveat in the nominal Treasury securities market justifies the U.S. Treasury Department's decision to launch the TIPS program.

[4] The literature on portfolio constraints affecting underinvestment in arbitrage includes agents' heterogeneity (Basak and Croitoru, 2000), wealth constraints (Loewenstein and Willard, 2000), and solvency constraints (Hindy, 1995).

2.2. Growing pains

TIPS do not have to compensate for the inflation risk, which positively affects TIPS prices relative to Treasury securities. However, the lack of liquidity in the early years results in lower TIPS prices and therefore higher borrowing costs relative to nominal Treasury issues. Eliminating inflation risk would countervail TIPS illiquidity premium during the program's early stages. Ex-post analysis shows that the illiquidity premium far outweighs the inflation risk premium; thus, TIPS are priced at relative discounts because of the lack of liquidity. The liquidity premium is estimated at approximately 30 basis points during normal times (Pflueger and Viceira, 2011). The gap between TIPS real yields and risk-free real yields in the early years is 100 basis points (d'Amico et al., 2018). TIPS-bearing liquidity premium are undervalued prior to 2004 and the illiquidity concern has led to TIPS being significantly discounted during the 2008 crisis (Haubrich et al., 2012). Consequently, TIPS in the early years are not cost-effective for issuers (Sack and Elsasser, 2004).

However, Fig. 1a shows that market liquidity of TIPS has dramatically improved. For instance, the daily trading volume of TIPS has more than doubled since the early stages of the program or the 2008 global financial crisis. The average daily trading volume of TIPS by primary dealers is $19.42 billion in 2020, $21.60 billion in 2021, and $18.59 billion in 2022, which is 1.23 %, 1.25 %, 0.97 % of the total TIPS outstanding at the end of each year, respectively.[5] Fig. 1b indicates that TIPS outstanding at the end of 2022 accounts for 7.97 % of the total marketable securities issued by the U.S. government. Moreover, illiquidity issues in the TIPS market have become immaterial. For example, liquidity premium and inflation risk premium are balanced after 2004 (Dudley et al., 2009) and liquidity premium is decreasing (d'Amico et al., 2018). Driessen et al. (2017) also document that liquidity correction only partially justifies TIPS underpricing, as 29.40 % (assuming market segmentation) or 62.06 % (assuming market integration) of TIPS discounts remain unexplained.

2.3. The TIPS–Treasury bond puzzle

Inflation swaps are a tradable form of BEI that exchanges realized inflation against a strike inflation rate.[6] In practice, inflation swap strikes are almost always higher than the BEI measures implied from nominal Treasury securities and TIPS (Fleming and Sporn, 2013). Fleckenstein et al. (2014) show that we can synthetically replicate nominal Treasury bonds by linearly combining TIPS, zero-coupon inflation swaps, and Separate Trading of Registered Interest and Principal of Securities (STRIPS). The replication strategy further confirms that synthetic Treasury bond prices are significantly lower than nominal Treasury bond prices. Fleckenstein et al. (2014) further document that taxation differences, inflation swap liquidity, repo financing, value of deflation floor embedded in TIPS, and convexity adjustment between nominal and real yield are unlikely to affect the mispricing.[7] Rather, they argue that TIPS discounts can be attributed to the fact that a violation of the no-arbitrage condition may not immediately disappear, possibly because capital does not flow instantaneously (Mitchell et al., 2007; Duffie, 2010; Mitchell and Pulvino, 2012). Nonetheless, TIPS discounts are still substantial. Fig. 2 displays that inflation swaps are still considerably higher than BEI measures, indicating that price gaps do not disappear but rather exist over a prolonged horizon.

2.4. Potential determinants of the TIPS–Treasury bond puzzle

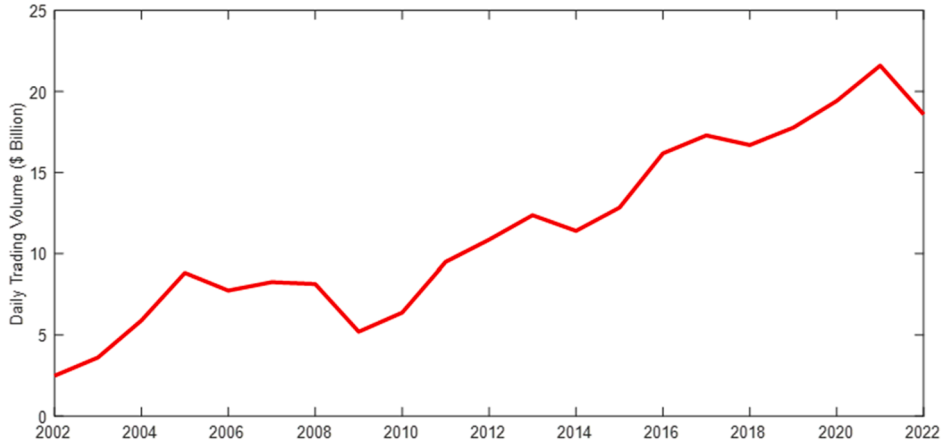
TIPS discounts are persistent and still sizable. Understanding the factors associated with the expansion and contraction of the basis over the last two decades is essential. We note that Treasury securities are primarily traded in OTC markets in which intermediaries play a key role (Allen, 2001). Thus, institutional idiosyncrasies could influence inflation-indexed bond yields, as discussed in Campbell et al. (2009a). For instance, the slow development of indirect investment vehicles investing in TIPS, such as mutual funds, could have contributed to high TIPS yields in the early stages, showing a link to the lack of liquidity and slow adjustment to a new type of financial instrument. The low yield in the U.K. inflation linkers market is probably owing to strong demand by U.K. pension funds, representing the demand and supply effects in inflation-indexed bond markets (Greenwood and Vayanos, 2014). A liquidity episode caused by the collapse of Lehman Brothers resulted in anomalies in the TIPS market and extreme volatility in TIPS yields, as the unwinding of institutional positions created severe circumstances.

We conjecture that the unique characteristics of financial intermediation could be potential sources of TIPS discounts. This direction is consistent with intermediary asset pricing theories, in which intermediaries' risk-bearing capacity matters (He and

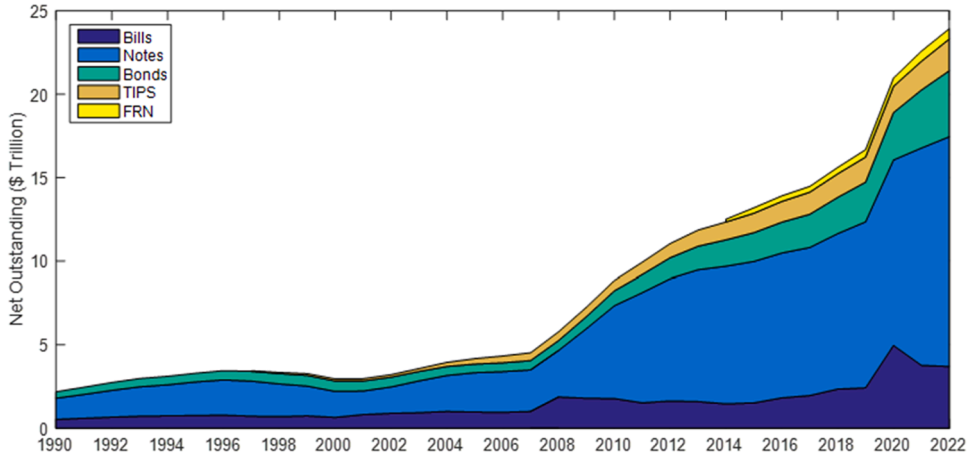
[5] According to Securities Industry and Financial Markets Association, the average daily trading volume of Treasury coupon securities (i.e., Treasury notes and bonds) is $409.75 billion in 2020, $446.77 billion in 2021, and $438.14 billion in 2022, which is 2.94%, 2.71%, 2.47% of total outstanding at the end of 2020, 2021, and 2022, respectively. For Treasury bills, the average is $170.85 billion in 2020, $152.85 billion in 2021, and $155.07 billion in 2022, which is 3.44%, 4.05%, and 4.19% at the end of 2020, 2021, and 2022, respectively.

[6] With the presence of inflation swaps, it is argued that the combined liquidity premium in inflation swaps and TIPS results in TIPS discounts (Christensen and Gillan, 2012). Alternatively, Haubrich et al. (2012) conclude that the introduction of inflation swaps in 2003 eliminated TIPS mispricing from 2004 to mid-2008.

[7] Deflation floor represents that TIPS' redemption price at maturity is the higher of face value or the inflation-adjusted notional amount. In case of deflation, this provision protects investors' redemption at face value of TIPS. This safeguard against deflation is only applicable to the principal repayment at maturity. Even though the deflation floor would only be triggered for extreme cases (i.e., deep out-of-the-money option), such as the Great Recession in the 1930s, the deflator clearly has a non-zero positive value. Excluding the deflator at maturity marginally decreases the TIPS prices, and thus widens the price gaps between TIPS and Treasury bonds. Our results rest on TIPS with embedded deflator.



(a) TIPS trading volume



(b) TIPS notional debt outstanding

Fig. 1. TIPS trading volume and outstanding notional.
Fig. 1a represents the average daily trading volume of TIPS by primary dealers, and Fig. 1b illustrates the outstanding notional of U.S. Treasury marketable securities. The data are reported by the Federal Reserve Bank of New York and the statistics are tabulated by the Securities Industry and Financial Markets Association.

Krishnamurthy, 2013; He et al., 2017; Adrian and Shin, 2014). Specifically, the covariance risk between Treasury securities and equity has altered its signs over time (Campbell et al., 2009b) and intermediation frictions help explain the time-varying beta that represents inconvenience yield of Treasury securities (He et al., 2022).

Direct holding of securities and repo positions are subject to intermediaries' risk-bearing capacity. The intermediation frictions for direct holding of securities rely on the fact that dealers distinguish their pricing kernel for mimicking derivatives assets from physical assets (He et al., 2022). As such, intermediary frictions for direct holding of securities are measured as spreads between Treasury bonds and overnight index swaps (OIS). Repo financing and rehypothecation play a pivotal role in dealers' intermediation activities. The intermediation frictions for indirect holding of securities are associated with observations in which the general collateral repo rates (dealers lend to levered investors against collateral that will be kept in dealers' balance sheet) deviate from tri-party repo rates (dealers borrow money while providing collateral). Hence, the repo spreads measure intermediary frictions for their indirect holdings of securities (He et al., 2022). We predict that both proxies are positively associated with the magnitude of the TIPS mispricing, as they represent intermediaries' limited risk-bearing capacity.

On-the-run securities confer a special financing benefit to holders in the form of repo specialness (Duffie, 1996; Jordan and Jordan, 1997; Krishnamurthy, 2002; Banerjee and Graveline, 2013). Security-driven repo demand renders arbitrage challenging, as (relatively expensive) Treasury bonds are to be sourced in the repo market and then short-sold to purse textbook arbitrage profits from TIPS mispricing. Moreover, intermediaries' capital and margin requirements are associated with market liquidity (Brunnermeier and Pedersen, 2009), affecting asset prices in the financial market (Acharya and Pedersen, 2005). When capital and margin requirements

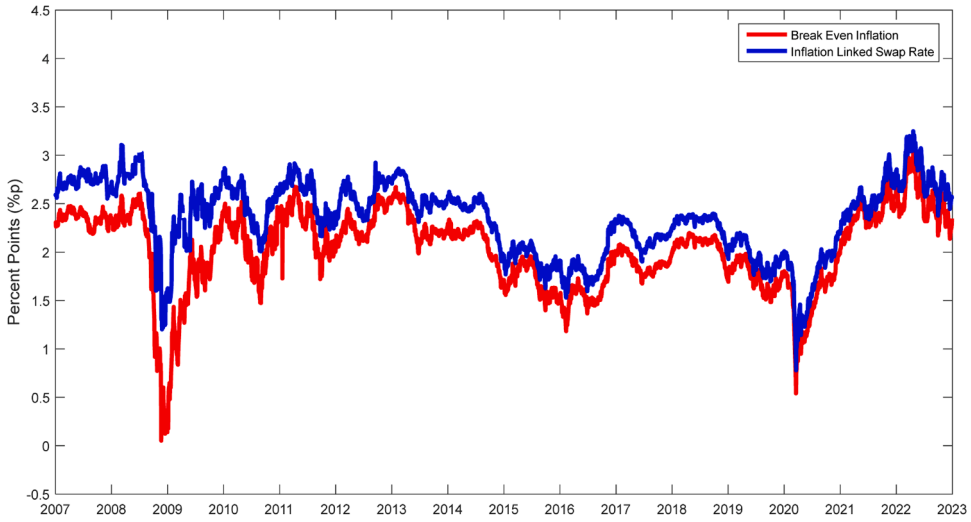


Fig. 2. Market-based inflation measures (BEI vs. inflation-linked swap rate).
Fig. 2 presents the two proxies for the market-based inflation measures. BEI refers to the differences between nominal and TIPS rates, in which nominal yields are 10-year constant maturity Treasury yields and TIPS yields are 10-year TIPS real yields. Inflation swap rates are the strike rates quoted in over-the-counter markets. Data are from J.P. Morgan and are tabulated daily. Data are from 2007 to 2022.

are binding, it is onerous for intermediaries to chase arbitrage profits owing to their margin constraints, even if such trades are free lunches in textbook sense (Garleanu and Pedersen, 2011).

We also consider strategic concerns, such as concerns about marking-to-market losses. Arbitrageurs may want to avoid volatile positions if the time-horizon is long. When TIPS discounts are implemented as arbitrage, total profit from the arbitrage is quantified at the initiation of trade. However, as the deal elapses until the matched maturity of TIPS and nominal bonds, unrealized profits and losses in the middle of the suggested investment horizon could significantly differ from the estimated total profits. For instance, if arbitrage opportunities occur with 10-year TIPS, it takes 10 years to fully realize the free lunch, with the possibility of experiencing marking-to-market losses at any time during the same period. Hence, future uncertainty amplifies fluctuations in marking-to-market profits and losses, even in textbook arbitrage deals, which makes arbitrage risky for constrained investors (Liu and Longstaff, 2004).[8] We utilize two ex-ante fixed-income uncertainty measures to represent constrained investors' strategic concerns regarding fluctuations in fixed income and relevant derivatives markets. The Merrill Lynch Option Volatility Estimate (MOVE) refers to the implied volatility from options on Treasury securities. The CBOE Interest Rate Volatility Index (SRVIX) by the Chicago Board Options Exchange (CBOE) represents swap market volatility, which is the fair strike of a variance swap contract implied from swaption prices (Britten-Jones and Neuberger, 2000; Bakshi and Madan, 2000; Jiang and Tian, 2005; Carr and Wu, 2009). Table 1 summarizes our proxies for the explanatory variables, as well as their construction and sources.

3. Quantifying the TIPS–Treasury bond mispricing

We reiterate that synthetic Treasury bonds can be replicated via a linear combination of TIPS, zero-coupon inflation swaps, and STRIPS (Fleckenstein et al., 2014). However, this approach requires a basket of inflation swaps and STRIPS maturing at every coupon payment date and maturity date. We can achieve the same results through an asset swap contract without relying on plenty of micro positions in inflation swaps and STRIPS. We believe that utilizing asset swaps rather than numerous inflation swaps and STRIPS is methodologically simpler and more practical.

3.1. Replicating a synthetic bond

We recap Fleckenstein et al. (2014) methodology and describe how our asset swap approach can simplify the process. To gage the degree of mispricing in the TIPS market, Fleckenstein et al. (2014) replicate a synthetic bond whose payoffs are identical to those of a nominal Treasury bond. As TIPS payments reflect realized inflation measured by the CPI, two pivotal points are considered for a proper comparison between TIPS and the corresponding Treasury bonds. First, realized inflation in TIPS must be substituted with an expected inflation measure. The inception of inflation-linked derivatives in the early 2000s is critical because inflation swaps make it possible to exchange realized inflation in TIPS payoffs with expected inflation, or vice versa. Second, inflation-swapped TIPS coupons and

[8] Ex-ante uncertainties also have a link to liquidity. Overall uncertainty has a significant market-wide impact on individual asset' liquidity and dominate the combined effects of other influential determinants (Chung and Chuwonganant, 2014).

Table 1
Empirical proxies for the potential drivers of the TIPS–Treasury bond puzzle.

Category	Driver	Proxies	Timespan	Sources
Intermediation frictions	Intermediaries' direct book cost	10-year Treasury minus 10-year OIS	Jan. 2005–Dec. 2022	J.P. Morgan
	Intermediaries' indirect book cost	GC repo rate minus tri-party repo rate	Aug. 2012–Dec. 2022	GC repo rates from DTCC and tri-party repo rates from Bank of New York Mellon
	Funding liquidity	3-month LIBOR minus 3-month T-bill	Jan. 2005–Dec. 2022	J.P. Morgan
	Repo specialness	GC repo rate minus On-the-run Treasury Special Repo Rate	Jan. 2005–Dec. 2022	GC repo rates from DTCC and special repo rates from J.P. Morgan
Strategic concerns	Ex-ante bond option volatility	MOVE index	Jan. 2005–Dec. 2022	Bloomberg
	Ex-ante swap rate volatility	SRVIX index	Jan. 2005–Dec. 2022	Own calculation using swaption data provided by J.P. Morgan

This table presents the drivers of the TIPS–Treasury bond puzzle. Proxies are classified into two categories: a) intermediation friction and b) strategic concerns. The table below includes various abbreviations defined as follows: OIS = overnight index swap, GC repo = general collateral repo rates, LIBOR = London Interbank Offered Rate, and DTCC = the Depository Trust and Clearing Corporation. DTCC GC repo rates are available at https://www.dtcc.com/charts/dtcc-gcf-repo-index#download and the Bank of New York Mellon tri-party repo rates are from https://repoindex.bnymellon.com/repoindex/. We follow CBOE's SRVIX white paper to replicate the index. Replication resources are available at https://cdn.cboe.com/resources/indices/srvix-white-paper.pdf.

principal payments must be matched precisely with the corresponding nominal Treasury bond coupons and principal using STRIPS. Consequently, inflation-swapped and STRIPS-matched coupons and principal payment of TIPS reproduce nominal Treasury securities.[9]

Table 2 summarizes the strategy for creating a synthetic Treasury bond consisting of TIPS, inflation swaps, and STRIPS. Given that the suggested relationship is a linear combination of available assets, the aggregate position of synthetic TIPS must be indistinguishable from corresponding nominal Treasuries in a competitive equilibrium. However, nominal Treasuries are usually more expensive than synthetic Treasuries (Fleckenstein et al., 2014). Thus, a replicating strategy in which traders take a long synthetic TIPS position against shorting a nominal Treasury indicates a textbook arbitrage opportunity. Thus, the wedge between two U.S. government securities may not be characterized by a traditional rational pricing paradigm. This mispricing clearly violates the law of one price.

Despite its intuitive appeal, Fleckenstein et al. (2014) replication strategy requires a basket of micro positions on inflation swaps and STRIPS to match coupons and notional precisely. For instance, investors who try to replicate a 10-year Treasury bond must trade 20 contracts of zero-coupon inflation swaps and 20 lots of STRIPS whose maturities coincide with the semi-annual coupon payment schedule of the target bond. Trading such micro positions might be onerous and costly because the size of inflation swaps differs from the regular trading size per contract.

3.2. *Asset swaps and synthetic floaters*

Different from Fleckenstein et al. (2014), we propose a much simpler approach of using asset swaps in OTC markets.[10,11] Rather than establishing a synthetic Treasury bond, we transform TIPS and nominal Treasuries into two identical floaters via asset swaps. Recall that asset swaps are OTC credit derivative contracts that create synthetic floating-rate notes. While purchasing a bond, an asset swap buyer enters into a tailored interest rate swap (IRS) in which the asset swap buyer pays a fixed coupon equivalent to the coupon rate of the bond in exchange for the London Interbank Offered Rate (LIBOR) plus asset swap spreads. As such, asset swaps are customized IRS, in which the fixed leg (i.e., the asset swap buyer) is set to be equal to the coupons and principal of particular reference bonds, and the floating leg (i.e., the asset swap seller) pays LIBOR plus spreads mirroring credit risks that the fixed payer takes. If the bond defaults, the asset swap buyer still has to honor the fixed payments that are not funded with the cash flows of the reference bond anymore (Duffie, 1999; Aussenegg et al., 2016). Given that asset swap buyers take on the default risk, the asset swap spreads are considered economic compensation for the default-contingent exposure, offering credit protection to the asset swap sellers. Thus, the asset swap spreads (i.e., protection fees) measure the embedded credit risk of a bond in the form of spreads over the LIBOR.[12]

[9] Andreasen et al. (2021) argue that relying on inflation swaps is debatable due to poor liquidity in inflation swaps. However, a dealer survey in April 2012 shows that indicative bid-ask spreads of inflation swaps are shown to be 2 and 3 basis points, realized bid-ask spreads are reported to be 2.8 basis points, and the overall market size is thought to be around $350 million per day (Fleming and Sporn, 2013). Fleckenstein et al.'s (2014) results are also not associated with distortions in the inflation swap market, such as illiquidity, hedging costs, and counterparty credit risk.

[10] Fleckenstein et al. (2014) also hint on an alternative way of implementing an arbitrage using asset swaps. For details, see their Footnote 11. Besides, inflation swaps and asset swaps are complements, as Campbell et al. (2009a) state that investors hedge their inflation swap positions in the asset swap market.

[11] In practice, banks widely utilize asset swaps to manage their balance sheet. Asset swaps convert banks' long-term fixed-rate assets to short-term assets. Consequently, asset swaps enable banks to match asset duration with short-term liabilities, such as deposits.

[12] Consequently, asset swaps are widely recognized as credit derivatives because asset swap spreads quantify the reference bond' credit risk.

Table 2
Mechanism for replicating a synthetic bond.

	Nominal Treasury			Synthetic Treasury		
	Upfront at inception	Semi-annual payments	At maturity	Upfront at inception	Semi-annual payments	At maturity
Treasury	$-P$	$+C$	$C+100$			
TIPS				$-V$	$+R\cdot I_t$	$+(R+100)\cdot I_T$
Inflation swap					$+R\cdot(K_t-I_t)$	$+(R+100)\cdot(K_T-I_T)$
STRIPS				$-\sum_{i=1}^{i=T}(C-R\cdot K_i)\cdot D_i-100\cdot(1-K_T)\cdot D_T$	$C-R\cdot K_t$	$(C-R\cdot K_T)+100\cdot(1-K_T)$
Net cash flows	$-P$	$+C$	$C+100$	$-\sum_{i=1}^{i=T}(C-R\cdot K_i)\cdot D_i-100\cdot(1-K_T)\cdot D_T-V$	$+C$	$C+100$

This table describes Fleckenstein et al. (2014) replication strategy to create a synthetic nominal Treasury. The key differences between nominal Treasuries and TIPS are two-fold. First, the face value of TIPS increases in accordance with the Urban CPI. Second, the TIPS coupon payments are adjusted by the CPI figures. P denotes the dirty price of a Treasury bond with a face value of 100, while paying C as coupons. Similarly, V represents the dirty price of a TIPS with a face value of 100 at issuance, while paying R as coupons. I_t equals the CPI index at time t divided by the initial CPI at time zero. A zero-coupon inflation swap buyer pays a compounded fixed rate K_t and receives I_t at time t. D_t represents the unit price of STRIPS maturing at time t.

For TIPS, asset swaps include the exchange of proceeds that comprise both the price of a bond and any accrued interest (i.e., dirty price), future coupon payments from an adjusted principal, and the final redemption. Because an asset swap contract transforms the entire cash flow of a bond into a floater, asset swap spreads represent a bond issuer's credit risk. Hence, an asset swap for TIPS is equivalent to a synthetic floating-rate note issued by the U.S. government. Similarly, the asset swap of a nominal Treasury is identical to that of another synthetic floating-rate note from the same issuer. All other things being equal, an asset swap spread of a TIPS issue and another asset swap spread from a nominal Treasury are identical because both spreads gage the credit quality of the same issuer. In addition, from the perspective of credit risk, TIPS and nominal Treasury securities are subject to the same provisions in the relevant credit markets. Thus, the difference between the two asset swap spreads measures the magnitude of relative mispricing in a simpler way. To summarize, asset swaps transform the price gaps between TIPS and nominal Treasury securities from a pure credit play to a comparable angle.

Obviously, U.S. government securities are also subject to default with positive credit default swap spreads in the credit market. Because TIPS pay coupons and the principal based on inflation-adjusted face values, cash flows from TIPS are concentrated on future dates relative to nominal Treasury securities. Given that the probability of default is non-zero, investors could consider nominal Treasury instruments safer than inflation-indexed securities whose effective durations are longer. Asset swaps effectively mitigate these concerns as cash flows are passed through to swap counterparties, eventually leaving typical floaters' cash flows with spreads over the LIBOR.[13] Table 3 illustrates the mechanics of asset swaps that create synthetic floaters from both nominal Treasuries and TIPS.[14]

Fig. 3 displays asset swap spreads of 10-year nominal Treasury bonds and TIPS, respectively. It shows that asset swap spreads of TIPS are almost always higher than those of nominal Treasury bonds, indicating the relative underpricing of TIPS to nominal Treasury securities from a different perspective. From 2005 to 2022, the gap between the two spreads is 31.80 basis points on average, peaking at 161.01 basis points during the 2008 crisis. The economic magnitude of TIPS discounts is approximately 3.18 % (31.80 basis points per annum × 10-year swap tenor) of asset swap notional on average while approaching 16.10 % (161.01 basis points per annum × 10-year swap tenor) of swap notional.[15] Fig. 3 further highlights the persistence of the price gaps, which calls for scholarly investigation.

Fleckenstein et al. (2014) compare synthetic Treasuries with nominal Treasuries. Hence, their arbitrage strategy is to take long positions on synthetic (i.e., inflation-swapped and STRIPS-matched) TIPS and short positions on nominal Treasury bonds. Contrastingly, we consider asset swaps to transform TIPS and nominal Treasury bonds into two synthetic floaters. Then, a strategy in which an investor simultaneously takes long positions on a relatively cheaper synthetic floater consisting of TIPS and an asset swap (therefore, receiving relatively higher asset swap spreads on TIPS) and short positions on another synthetic floater comprising a nominal Treasury bond and another asset swap (thus, paying relatively lower spreads on the nominal Treasury bonds) constitutes an arbitrage transaction to capture the designated spread differentials in asset swaps. To elaborate, the alternative strategy refers to purchasing a TIPS

[13] One could raise concerns on counterparty credit risks for asset swaps traded in OTC markets. However, Arora et al. (2012) show that the effect of counterparty credit risk on the pricing of credit default swap contracts is modest. Market participants in OTC markets require full collateralization of swap liabilities, leading to insignificant effects of counterparty credit risks on the pricing of swap contracts.

[14] A similar approach is employed in Ahn and Ahn (2022) who compare a synthetic floater (i.e., asset-swapped nominal Treasuries) with an actual floating rate note. Our quantification rests on a comparison between asset-swapped nominal Treasuries and TIPS.

[15] Fleckenstein et al. (2014) report that TIPS discounts exceed $20 per $100 notional in extreme cases. For the sake of simplicity, we disregard the time value of the spread differentials.

Table 3
Mechanism of asset swaps.

	Asset-swapped nominal Treasury			Asset-swapped inflation linker		
	Upfront at inception	Semi-annual payments	At maturity	Upfront at inception	Semi-annual payments	At maturity
Treasury	$-P$	$+C$	$C+100$			
TIPS				$-V$	$+R\cdot I_t$	$+(R+100)\cdot I_T$
Asset swap (fixed)	$+P$	$-C$	$-(C+100)$	$+V$	$-R\cdot I_t$	$-(R+100)\cdot I_T$
Asset swap (floating)	$-P$	$+(LIBOR+N)\cdot P$	$+P$	$-V$	$+(LIBOR+L)\cdot V$	$+V$
Net cash flows	$-P$	$+(LIBOR+N)\cdot P$	$+P$	$-V$	$+(LIBOR+L)\cdot V$	$+V$

This table illustrates the mechanics of our approach, which creates two floating-rate notes using asset swaps. In the U.S., inflation linkers are typically traded as the proceeds of an asset swap, which adjusts its notion for accrued inflation. For comparability, nominal Treasuries are expressed in the form of a proceeds asset swap. P denotes the dirty price of a Treasury bond with a face value of 100, while paying C as coupons. Similarly, V represents the dirty price of a TIPS with a face value of 100 at issuance, while paying R as coupons. I_t equals the CPI index at time t divided by the initial CPI index at time zero. A zero-coupon inflation swap buyer pays a compounded fixed rate K_t and receives I_t at time t. LIBOR is the acronym of London Interbank Offered Rate. N refers to the asset swap spread for a nominal Treasury bond. L represents the asset swap spread for a TIPS issue.

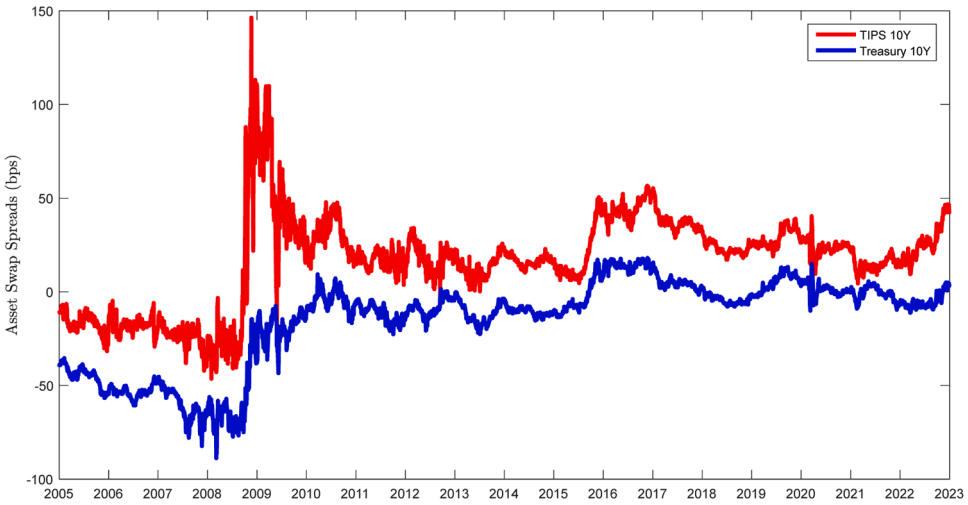


Fig. 3. Asset swap spreads.
Fig. 3 depicts the asset swap spreads of 10-year TIPS and maturity-matched nominal Treasury, which are added to the LIBOR rates. Asset swap spreads are provided by J. P. Morgan and are tabulated daily. The dataset covers the 2005–2022 period.

issue, taking on the asset swap buyer (pay fixed coupons and receive LIBOR plus asset swap spreads) on the TIPS issue, shorting a nominal Treasury bond, and taking on the asset swap seller (receive fixed coupons and pay LIBOR plus asset swap spreads) on the nominal Treasury bond. The fixed payer leg of the asset swap (i.e., the asset swap buyer) on TIPS is to be funded with the TIPS purchased. The obligations with nominal Treasury bond short-sold are to be sourced with the fixed receiver leg of an asset swap (i.e., the asset swap seller) on the nominal Treasury bond. Taken together, these series of transactions generate an arbitrage profit amounting to the asset swap spread differentials multiplied by the tenor of the swap.

4. Data and results

Empirical evidence that the TIPS–Treasury bond mispricing still exists in the financial market challenges our understanding of how the TIPS market works. The magnitude of the mispricing amounts to 3.18 % on average from 2005 to 2022. In this section, we demystify the factors that could drive the persistent TIPS–Treasury bond mispricing.

4.1. Data and descriptive statistics

We use J.P. Morgan's proprietary dataset to quantify the TIPS discount.[16] We obtain daily data from January. 3, 2005, to December. 30, 2022, but disregard the observations before the 2008 financial crisis. We aggregate the daily data from 2010 to 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies. Additionally, we winsorize the variables at the 1 % and 99 % levels to mitigate the impact of outliers on the results.

Table 4 presents summary statistics for our key variables, while Table 5 presents a correlation matrix between the key variables and TIPS discounts. Table 5 shows that the key variables are highly intertwined with each other. This complex covariance structure should be addressed in the remaining statistical analyses.

4.2. Univariate regression analyses

We first employ simple ordinary least squares univariate regression analyses to confirm the link from each potential driver to TIPS discounts.

$$Y_t = X_t\beta_1 + C_t\gamma_1 + \varepsilon_t \tag{1}$$

where the time interval is weekly; $\boldsymbol{Y}$ is a column vector that denotes TIPS discounts or premium in comparison to Treasury bonds; $\boldsymbol{X}$ represents each of the potential drivers of TIPS discounts; $\boldsymbol{C}$ is a control variable matrix; and ε is a white noise vector. The control variables include the inflation swap rate, the return on the S&P 500 index, the CBOE Volatility Index (VIX), and Moody's seasoned Baa corporate bond yield relative to the yield on the 10-year Treasury constant maturity.[17] We linearly scale both the explanatory and dependent variables in Eq. (1). Thus, any pair of coefficients can be compared to one another for the purpose of judging which explanatory variable has a relatively larger impact on the TIPS–Treasury bond mispricing.

Table 6 reports the univariate regression analysis results for the TIPS discounts in Eq. (1). All key variables are significantly associated with TIPS deviations at the 1 % level, and the F statistics indicate that the regression analysis results are significant. After controlling for the inflation swap rate, the return on the S&P 500 index, VIX, and Moody's seasoned Baa corporate bond yield relative to yield on 10-year Treasury constant maturity, the association between each key variable and TIPS deviations remains significantly positive. As frictions (i.e., intermediaries' direct and indirect risk-bearing capacity, funding and margin constraints, huddles to short-selling) increase, investors are less attracted to capture the TIPS mispricing. Moreover, investors are not drawn toward arbitrage opportunities when they are concerned about escalated future uncertainty. Specifically, a one standard deviation increase in each of TIY (direct holdings), TIY (indirect holdings), MOVE, SRVIX, TED, and repo specialness, leads to a standard deviation increase of 0.4145, 0.3863, 0.3250, 0.7682, 0.3324, and 0.1352 in the TIPS–Treasury mispricing, respectively.[18] Increasing SRVIX has approximately twice to six times the impact on the mispricing as increasing the other variables.

The univariate regression analysis in Eq. (1) determines whether each potential driver is significantly associated with TIPS discounts on the margin after control variables are considered. In Table 7, we conduct a more stringent test. We consider year-month fixed effects and further test whether the empirical association documented in Table 6 does not vanish. The univariate fixed-effects regression analysis results show that a one standard deviation increase in SRVIX results in a 0.5386 standard deviation increase of the TIPS–Treasury bond mispricing. TED and repo specialness are no longer significantly associated with mispricing.

In the last columns of Tables 6 and 7, we horse-race the six explanatory variables. We confirm that those explanatory variables that denote market uncertainty or intermediation frictions win the race. A one standard deviation increase in market uncertainty (TIY for direct holdings of securities) leads to a 0.4764 (0.3584) standard deviation increase in the mispricing. The results in Tables 6 and 7 suggest that uncertainty-related issues in the fixed-income market could be the key to the understanding of the mispricing.

4.3. Principal components analysis and multivariate results

We have seen in Table 5 that the potential drivers of the TIPS–Treasury bond puzzle are highly correlated. Therefore, we utilize both principal component analysis and feature selection techniques to address multicollinearity issues. We first check whether there exists a low-dimensional (and linearly independent) representation of the key variable space and then proceed to see whether this new representation can capture as much information as possible. Using TIY (direct holdings), TIY (indirect holdings), MOVE, SRVIX, TED, and repo specialness, we extract the principal components of the correlation matrix. To avoid potential bias, we randomly select 50 %

[16] We analyze 10-year TIPS. A 10-year TIPS is the only security introduced in the first auction in January 1997 and continues to be auctioned without any discontinuation and reintroduction. The other tenors, such as 5-, 20-, or 30-year, are introduced later, discontinued (and oftentimes reintroduced), or reduced to infrequent issuance schedule. For example, 5-year tenor is first introduced in June 1997, discontinued in 1998, and reintroduced in October 2004. A detailed timeline is available at https://www.treasurydirect.gov/research-center/timeline/tips/.

[17] We obtain the inflation swap data, whose maturity is matched to TIPS maturity, from J.P. Morgan. The other control variables data can be downloaded on the Federal Reserve Economic Data website.

[18] TIY (direct holdings) represents intermediaries' book costs for the direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus the tri-party repo rate. MOVE represents the implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rate minus on-the-run Treasury special repo rate.

Table 4
Summary statistics.

	n	Mean	SD.	Min	P25	Median	P75	Max
Deviation	677	8.32	2.61	3.10	6.56	8.02	9.70	19.49
TIY (direct holdings)	665	7.78	3.50	2.38	5.27	7.01	9.66	19.64
TIY (indirect holdings)	536	2.23	1.99	−1.35	0.97	1.95	3.16	9.51
MOVE	635	23.01	7.96	12.32	17.01	21.54	26.86	64.32
SRVIX	677	29.11	7.48	19.10	23.57	26.99	33.37	59.82
TED	669	8.56	4.71	2.33	5.44	7.37	10.66	43.24
Repo specialness	669	9.12	14.11	−2.30	−2.23	4.70	9.58	96.80

We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. We winsorize the variables at the 1 % and 99 % levels to mitigate the impact of outliers on the results. n denotes the number of observations. SD denotes the standard deviation. Deviation refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. TIY (direct holdings) represents intermediaries' book costs for the direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus the tri-party repo rate. MOVE represents the implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rate minus on-the-run Treasury special repo rate.

Table 5
Correlation coefficients.

	Deviation	TIY (direct holdings)	TIY (indirect holdings)	MOVE	SRVIX	TED	Repo specialness
Deviation	1.0000	0.4515	0.2894	0.5401	0.6433	0.4360	0.1281
		[< 0.0001]	[< 0.0001]	[< 0.0001]	[< 0.0001]	[< 0.0001]	[.0009]
TIY (direct holdings)		1.0000	0.6009	0.0670	0.0850	0.4853	0.2854
			[< 0.0001]	[.0947]	[.0284]	[< 0.0001]	[< 0.0001]
TIY (indirect holdings)			1.0000	−0.1446	−0.0584	0.3593	0.3157
				[.0012]	[.1765]	[< 0.0001]	[< 0.0001]
MOVE				1.0000	0.8909	0.3365	−0.0134
					[< 0.0001]	[< 0.0001]	[.7358]
SRVIX					1.0000	0.2916	−0.0402
						[.0001]	[.2973]
TED						1.0000	0.1129
							[.0035]
Repo specialness							1.0000

We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. P-values are shown in square brackets. Deviation refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. TIY (direct holdings) represents intermediaries' book costs for direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus the tri-party repo rate. MOVE represents the implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rate minus on-the-run Treasury special repo rate.

of the entire observations, calculate the covariance matrix of the chosen subsample, go over the process 10,000 times, and finally choose the covariance matrix with the smallest determinant. We use one as the threshold of eigenvalues. Table 8 shows that the first two principal components satisfy this threshold value criterion and collectively capture 69.79 % of the total variation.

Table 9 further shows how much and in which direction each potential driver loads on the two principal components reported in Table 8. To interpret each factor, we apply a varimax rotation of the two factors. Values in Table 8 are rounded off to the nearest integer after being multiplied by 100. We place an asterisk on those values greater than 0.50.

The varimax-rotated factor patterns in Table 9 reveal that Factor 1 is heavily loaded on MOVE and SRIVX; thus, it could denote marking-to-market concerns or liquidity risk for TIPS (Andreasen et al., 2021). Factor 2 is heavily loaded on TIY (direct holdings), TIY (indirect holdings), TED, and repo specialness; thus, it can capture intermediation costs in the TIPS market. We term the first and second principal components as "marking-to-market concerns" and "intermediation frictions," respectively, and use the two linearly independent principal components in the multivariate regression analyses instead of the six variables in Table 2. To investigate the impact of each of the two potential channels on TIPS discounts, we first standardize the key variables and linearly combine them into marking-to-market concerns and intermediation frictions. The low-dimensional proxies that denote marking-to-market concerns and intermediation frictions are normalized using their sample averages and standard deviations, allowing for the direct comparison of the magnitude of the estimated coefficients.

We use the two orthogonal factors (i.e., marking-to-market concerns and intermediation frictions) of the potential explanatory variables space and check which factor is empirically more correlated with TIPS discounts. Table 10 reports the results. We confirm that both marking-to-market concerns and intermediation frictions are significantly and positively associated with TIPS discounts at the 1 % level. Because the two variables of interest as well as the dependent variable are standardized, we conduct a meta-analysis to gage the relative importance of the variables. Following Ahn (2022) and De Caigny et al. (2020), we compare the absolute values of the

Table 6
Regression analyses with controls.

	Dependent variable: *Deviation*						
TIY (direct holdings)	0.4145***						0.3101***
	[15.34]						[8.03]
TIY (indirect holdings)		0.3863***					0.1391***
		[10.99]					[3.18]
MOVE			0.3250***				−0.3253***
			[7.67]				[−4.00]
SRVIX				0.7682***			0.9534***
				[15.50]			[8.81]
TED					0.3324***		0.1004***
					[5.83]		[2.93]
Repo specialness						0.1352***	0.0327
						[4.08]	[1.14]
Inflation swap rate	0.3911***	0.4982***	0.2356***	−0.5960***	0.3972***	0.3704***	0.0549
	[11.01]	[13.40]	[5.89]	[−6.77]	[10.10]	[9.89]	[1.28]
Return on the S&P500 index	0.0181	−0.0091	0.0258	0.0038	0.0348	0.0151	−0.0059
	[0.68]	[−0.35]	[1.13]	[0.16]	[0.97]	[0.61]	[−0.26]
VIX	0.1422***	0.1149***	−0.0486	−0.0670*	0.0183	0.0091**	0.0463
	[3.92]	[2.84]	[−1.18]	[−1.81]	[0.40]	[2.41]	[1.23]
BAA10Y-CMT10Y	0.1542***	0.0335	0.2429***	−0.0209	0.1664***	0.2645***	−0.2419***
	[4.02]	[0.76]	[6.07]	[−0.51]	[4.13]	[6.45]	[−5.37]
N	678	533	631	672	664	671	481
F statistic	127.86***	67.87***	79.04***	109.46***	95.80***	71.90***	80.68***
R^2	0.4943	0.3917	0.3874	0.4511	0.4213	0.3509	0.6319

This table summarizes the univariate regression analysis results for the deviation in Eq. (1). Deviation, measured in basis points, refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. TIY (direct holdings) represents intermediaries' book costs for direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus tri-party repo rate. MOVE represents implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rate minus on-the-run Treasury special repo rate. Controls include the inflation swap rate (%), the return on the S&P 500 index (%), VIX, and Moody's seasoned Baa corporate bond yield relative to the yield on the 10-year Treasury constant maturity (BAA10Y-CMT10Y,%). All variables are standardized. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. N denotes the number of observations. Heteroscedasticity-consistent test statistics are shown in square brackets. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

test statistics and the magnitude of the estimated coefficients for each variable of interest. In the multivariable regression analysis setting, the coefficient estimates (test statistics) for marking-to-market concerns and intermediation frictions are 0.3716 and 0.1199 (6.51 and 3.09), respectively. The significance of marking-to-market concerns is not subsumed in multivariate regression analyses. Hence, we posit that marking-to-market concerns are likely the key channel for inducing TIPS discounts; although, the impact of intermediation friction remains substantial.

4.4. Selecting key features: the elastic net method

Using recent developments in machine learning, we conduct a more formal test to gage the relative importance of the factors associated with TIPS discounts and the extent of the association. Rapach et al. (2013) utilize feature selection techniques to quantify the relative predictability of U.S. returns in the global financial market. Chinco et al. (2019) also take advantage of variable selection techniques to identify key predictors when signals are sparse in the stock market. Following the literature, we employ the elastic net (Zhou and Hastie, 2005; Zhou and Zhang, 2009; Ghosh, 2011) to disentangle the relative explanatory power of marking-to-market concerns and intermediation frictions.[19]

The elastic net assumes two penalty terms, which can be written in the following Lagrangian form:

[19] The elastic net method is more robust than other feature selection techniques, for example back or forward stepwise regression. See Bai and Ng (2008) for further details. We do not consider the least absolute shrinkage and selection operator (LASSO) (Tibshirani, 1996). LASSO performs variable selection and regularization using an $AptCommandmathcall_1$ penalty term. The LASSO coefficients minimize the sum of the squared residuals subject to the $AptCommandmathcall_1$ penalty term as follows: $\sum_{i=1}^{n}[y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}]^2 + \lambda\sum_{j=1}^{p}|\beta_j|$. When λ is large enough, the $AptCommandmathcall_1$ penalty term forces less important or negligible coefficients to degenerate during the optimization process. LASSO works well when the number of variables exceeds the number of observations; although, in this case, the LASSO criteria are not strictly convex, implying that a unique solution is not guaranteed (e.g., Tibshirani, 2013). When two or more features contain identical information, LASSO's objective function fails to find a unique solution. This is because LASSO does not consider the correlation structure of the explanatory variable set. If the explanatory variables are highly intertwined in the data, LASSO selects only one variable. The elastic net method is designed to overcome the caveat inherited in LASSO when a group of variables is highly correlated. The elastic net method is indifferent from LASSO when the explanatory variables are orthogonal to each other.

Table 7
Univariate regression analyses with fixed effects.

	Dependent variable: *Deviation*						
TIY (direct holdings)	0.1542***						0.3584***
	[3.21]						[7.08]
TIY (indirect holdings)		0.0501*					0.0561**
		[1.89]					[2.17]
MOVE			0.1821***				−0.0205
			[3.49]				[−0.26]
SRVIX				0.5386***			0.4764***
				[9.68]			[3.70]
TED					0.0253		−0.0506
					[0.65]		[−1.57]
Repo specialness						−0.0171	−0.0043
						[−1.09]	[−0.32]
Inflation swap rate	0.3876***	0.5412***	0.3704***	0.3914*	0.4469***	0.4370***	0.1738***
	[8.42]	[13.09]	[7.46]	[1.93]	[9.67]	[9.76]	[2.79]
Return on the S&P500 index	−0.0165	−0.0101	−0.0209	−0.0146	−0.0193	−0.0197	0.0033
	[−1.22]	[−0.76]	[−1.49]	[−1.08]	[−1.40]	[−1.45]	[0.27]
VIX	0.0929**	0.0455	0.0275	0.0180	0.0851**	0.0902**	0.0299
	[2.35]	[1.21]	[0.62]	[0.44]	[2.10]	[2.26]	[0.76]
BAA10Y-CMT10Y	0.2487***	0.1766***	0.2573***	0.2516***	0.2664***	0.2864***	−0.0669
	[4.33]	[3.23]	[4.37]	[4.33]	[4.30]	[5.10]	[−1.06]
Year-month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	678	533	631	672	664	671	481
F statistic	45.20***	60.59***	42.06***	45.99***	44.85***	45.30***	68.40***
R^2	0.9354	0.9514	0.9347	0.9350	0.9345	0.9342	0.9639

This table presents univariate regression analysis results with year-month fixed effects. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. Deviation, measured in basis points, refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. TIY (direct holdings) represents intermediaries' book costs for direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus tri-party repo rate. MOVE represents implied Treasury bond option volatility. SRVIX proxies ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rates minus on-the-run Treasury special repo rates. All variables are standardized. N denotes the number of observations. Heteroscedasticity-consistent test statistics are shown in square brackets. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 8
Eigenvalues of the correlation matrix.

	Eigenvalue	Difference	Proportion	Cumulative
1	2.3378	0.4882	0.3896	0.3896
2	1.8496	0.9967	0.3083	0.6979
3	0.8529	0.3649	0.1422	0.8400
4	0.4880	0.1204	0.0813	0.9214
5	0.3676	0.2635	0.0613	0.9826
6	0.1041		0.0174	1.0000

This table presents the results of the principal component analysis. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. We consider the following six variables: TIY (direct holdings), TIY (indirect holdings), MOVE, SRVIX, TED, and repo specialness. TIY (direct holdings) represents intermediaries' book costs for the direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes intermediaries' indirect book costs and is measured as GC repo rate minus tri-party repo rate. MOVE represents the implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rates minus on-the-run Treasury special repo rate.

$$\sum_{i=1}^{n}\left[y_i-\beta_0-\sum_{j=1}^{p}\beta_j x_{ij}\right]^2+\lambda_1\sum_{j=1}^{p}|\beta_j|+\lambda_2\sum_{j=1}^{p}|\beta_j|^2 \tag{2}$$

We do not have strong priors on λ_1 and λ_2. We consider a 10,000 × 10,000 grid of $(\lambda_1,\ \lambda_2)$ values[20] and choose $(\lambda_1^*,\lambda_2^*)$ that yield the best criterion value according to Schwarz's Bayesian information criterion (Schwarz, 1978). The chosen tuning parameter values are relatively small, and the criterion surface is very flat near the chosen parameter values. Hence, there exist a wide range of $(\lambda_1,\ \lambda_2)$

[20] $(\lambda_1,\ \lambda_2)$ is defined over $[0,1]\times[0,1]$.⟨/*END*⟩We also try Akaike's information and obtain the same results.

Table 9
Varimax-rotated factor patterns.

	Factor 1	Factor 2
TIY (direct holdings)	19	83*
TIY (indirect holdings)	−16	85*
MOVE	96*	−3
SRVIX	95*	0
TED	48	58*
Repo specialness	−5	57*

This table describes the principal components of the varimax-rotated covariance matrix for TIY (direct holdings), TIY (indirect holdings), MOVE, SRVIX, TED, and repo specialness. TIY (direct holdings) represents intermediaries' book costs for the direct holding of securities and is measured as 10-year Treasury minus 10-year OIS rates. TIY (indirect holdings) denotes the intermediaries' indirect book costs and is measured as GC repo rates minus tri-party repo rates. MOVE represents the implied Treasury bond option volatility. SRVIX proxies for ex-ante swap market volatility implied from swaption prices. TED is 3-month LIBOR minus 3-month T-bill. Repo specialness is GC repo rates minus on-the-run Treasury special repo rates. We use one as the threshold of eigenvalues. The first two principal components satisfy this threshold value criterion. The values are rounded to the nearest integer after multiplying by 100. An asterisk is placed on values greater than 0.5.

Table 10
Multivariate regression analyses.

	Dependent variable: Deviation		
Marking-to-market concerns	0.2777***		0.3716***
	[4.30]		[6.51]
Intermediation frictions		0.1211***	0.1199***
		[3.20]	[3.09]
Inflation swap rate	0.3058***	0.5206***	0.3406***
	[5.58]	[12.89]	[7.02]
Return on the S&P500 index	−0.0210	−0.0080	−0.0085
	[−1.50]	[−0.61]	[−0.65]
VIX	0.0209	0.0449	−0.0556
	[.48]	[1.21]	[−1.41]
BAA10Y-CMT10Y	0.2189***	0.1426**	0.0716
	[3.64]	[2.54]	[1.24]
Year-month fixed effects	Yes	Yes	Yes
N	631	521	481
F statistic	42.65***	60.74***	60.15***
R^2	0.9355	0.9529	0.9575

This table summarizes the multivariate regression analysis results for the TIPS discounts. Deviation, measured in basis points, refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. Controls include the inflation swap rate, the return on the SP 500 index, the VIX, and Moody's seasoned Baa corporate bond yield relative to the yield on the 10-year Treasury constant maturity (BAA10Y-CMT10Y). We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. N denotes the number of observations. The heteroscedasticity-consistent test statistics are shown in square brackets. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

values that show very similar results.[21] Table 11 presents the results in which λ_1 is 0.0010 and λ_2 is 0.0015. Consistent with the results of the regression analyses in Table 10, the elastic net method finds that TIPS discounts are mainly associated with marking-to-market concerns, as the estimated coefficient for marking-to-market concerns is 0.5203 without controls.[22] In addition, the association between TIPS discounts and intermediation frictions is substantial in the data, as the estimated coefficient for intermediate frictions is 0.4048; although, the association is not as large as that of marking-to-market concerns. When considering the control variables, the inflation swap rate appears to be significant but the relative importance between marking-to-market concerns and intermediation frictions remains unchanged. Taken together, the results suggest that marking-to-market concerns are the main driver of TIPS discounts; though, intermediation frictions in the TIPS market also play a substantial role.

[21] We standardize all the variables. Hence, the magnitude of variables can be directly compared.

[22] If we see a curved pattern, we should consider a quadratic term. Multiple groupings of observations indicate that we might want to consider having classification effects in the estimation model.

Table 11
Feature selection techniques.

	Dependent variable: deviation	
Marking-to-market concerns	0.5203	0.4189
Intermediation frictions	0.4048	0.3203
Inflation swap rate		0.2643
Return on the S&P500 index		
VIX		
BAA10Y-CMT10Y		
Controls	No	Yes
Year-month fixed effects	No	No
N	484	484
F-statistic	245.30***	212.38***
R^2	0.5049	0.5714

This table summarizes the results from the elastic net method. Deviation, measured in basis points, refers to the 10-year TIPS asset swap spreads minus the 10-year Treasury asset swap spreads. Controls include the inflation swap rate, the return on the SP 500 index, the VIX, and Moody's seasoned Baa corporate bond yield relative to the yield on the 10-year Treasury constant maturity (BAA10Y-CMT10Y). All of the variables are normalized using sample mean and standard deviation. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noise and house-specific idiosyncrasies. N denotes the number of observations. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

4.5. Robustness of the results

We further implement the nonlinear iterative partial least squares (NIPALS) algorithm proposed by Wold (1975) to check the robustness of the results. We follow Wold (1994) and apply the variable-importance-in-projection (VIP) method to partial least squares (PLS) to obtain the cumulative measure of the influence of each potential determinant on TIPS discounts.

$$PLS-VIP \;=\; \left[K \times \frac{\sum_{i=1}^{7}\left(w_i^2 \times SSY_{comp,i}\right)}{SSY_{cum}}\right]^{1/2} \tag{3}$$

where w_i is the PLS weight; $SSY_{comp,i}$ is the sum of the squared discount (i.e., the Y variable) explained by component i; SSY_{cum} is the total sum of squares explained by PLS; and K is the number of terms in the model. Hence, the VIP is the weighted sum of the squares of

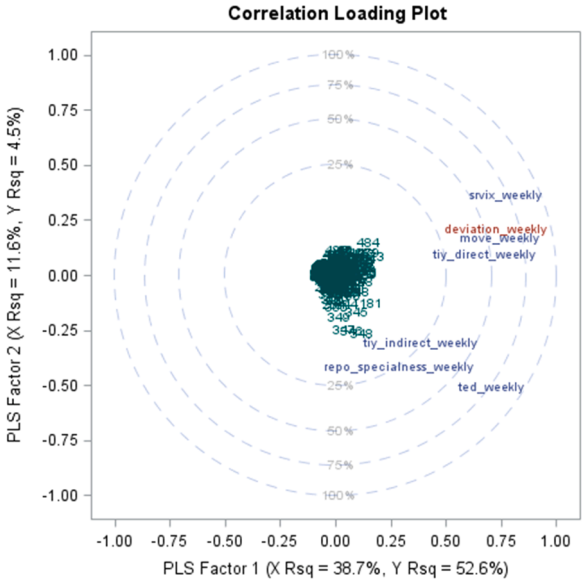


Fig. 4. Correlation loading plot.
Fig. 4 presents a correlation loading plot that summarizes the first two extracted factors, in which deviations represent TIPS discounts. We consider Treasury inconvenience yield (direct holdings), Treasury inconvenience yield (indirect holdings), MOVE, SRVIX, TED, and repo specialness. The dots in the plot represent the scores for the first two factors. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies.

all components of the PLS weight. Explanatory variables with VIP values greater than 1 are considered the most relevant, whereas those with VIP values less than 0.5 are considered irrelevant.

Fig. 4 presents a correlation loading plot summarizing the first two extracted PLS factors, in which deviations represent TIPS discounts. The correlation loading plot offers many useful aspects of the two most significant dimensions of the correlation matrix of the explanatory variables. The dots in the plot are highly centered and most of the observations are close together, implying that the first two factors can explain most variations.[23] PLS Factor 1′s dependent variable R^2 is 52.62 %; that is, the first PLS factor accounts for 52.62 % of the dependent variable's variation. The correlation loading plot also shows that the first PLS factor is heavily loaded on MOVE and SRVIX. Hence, we reassure that uncertainty-related issues in the fixed-income market could be the major drive of the TIPS–Treasury bond puzzle.

The variable importance plot provides more straightforward information regarding the drivers that are the most useful for predicting the dependent variable. The variable importance plot visualizes Wold's (1994) "variable importance for projection" statistics, which captures how each explanatory variable contributes to fitting the PLS for predictors and responses. Wold (1994) considers a value less than 0.8 to be "small" for the VIP. Fig. 5 shows that SRVIX, MOVE, and TIY (direct holdings) are the key to predicting TIPS discounts. Given that the three variables are the major building blocks of "marking-to-market concerns" and "intermediation frictions" in Table 9, the results further ascertain that both marking-to-market concerns and intermediation frictions play a key role to understand the TIPS–Treasury bond puzzle.

4.6. Time-varying covariance structure

The documented empirical associations could vary over time. We further employ Bai and Perron's (1998, 2003) test for structural breaks and explore the time-varying covariance structure between the potential channels and TIPS discounts. For $j = 1,\ldots,m+1$, we adopt the following specification:

$$Y_t = Z_t\delta_j + u_t \tag{4}$$

where m is the number of structural breaks; $\boldsymbol{Y}$ is a column vector representing the dependent variable; $\boldsymbol{Z}$ is the covariates vector; δ_j is the coefficients vector for each subsample; and u is the disturbance term. The $Z_t\delta_j$ term implies that all covariates are subject to structural changes. We identify the breaks by sequentially solving the following optimization problem for each $i = 1,\ldots,m$:

$$\sum_{i=1}^{m+1}\sum_{t=T_{i-1}+1}^{T_i}\left[Y_t - Z_t\delta_j\right]^2 \tag{5}$$

We begin with the single-break case. We should be able to determine the first structural break that minimizes the sum of squared residuals in Eq. (5). We repeat this process sequentially to search for further breaks. However, a caveat of this approach is that we do not have prior information on the exact number of structural breaks. Hence, we employ double maximum tests, in which the null hypothesis involves no break, and the alternative hypothesis involves an unknown number of breaks. The test statistic is:

$$D_{max} = \max_{n=1,\ldots,m}\left(a_n\ Sup\ F(0,n)\right) \tag{6}$$

where $F(0,n)$ denotes the standard F ratio test. If a_n is equal to 1 for all $n = 1,\ldots,m$, we conduct the UD_{max} test. Alternatively, if a_n is a function of the asymptotic critical values for $Sup\ F(0,n)$, the D_{max} statistic is called the WD_{max} test. In Table 12, both the UD_{max} and WD_{max} tests reject the null hypothesis at the 1% s level, implying multiple structural breaks. We sequentially identify at least two break days: January 15, 2021 and September 4, 2015. Table 12 summarizes the three regimes identified by the two structural break dates.

We run multivariate regression analyses with standardized variables to estimate the relative importance of the time-varying coefficients; that is, δ_j, for each subsample. Table 13 presents the results. The loadings on marking-to-market concerns and intermediation frictions are indeed time-varying, and this fluctuating covariance structure is associated with TIPS discounts from Regime 1 to Regime 3. Marking-to-market concerns are persistently priced throughout the data span. Moreover, the role of intermediation frictions is salient until the mid-2010s, and its impact on TIPS discounts has recently dwindled.

Because the results in Table 13 are dependent on an ex-post choice of the two relational shift dates, we are not free from the look-ahead bias. To address this concern, we allow both marking-to-market concerns and intermediation frictions to vary directly with macroeconomic factors. We consider the inflation swap rate and the effective Federal fund rate. By doing so, the interactive explanatory terms no longer rely on the ex-post information contents. Table 14 shows that the estimated coefficients on the interaction terms regarding intermediation frictions are positive and significant at the 1 % level. There exists a positive association between intermediation frictions and TIPS discounts over periods with high interest rates or high inflation. Marking-to-market concerns predominantly influence TIPS discounts. Thus, interacting the two potential channels with macroeconomic indicators conveys much the same economic insights as our main results in the previous sections.

In summary, unlike other Treasury securities in which the face value is fixed over time, the principal of a TIPS issue increases (or decreases in the case of deflation) in accordance with economic inflation. Although TIPS are designed to protect investors against

[23] If we see a curved pattern, we should consider a quadratic term. Multiple groupings of observations indicate that we might want to consider having classification effects in the estimation model.

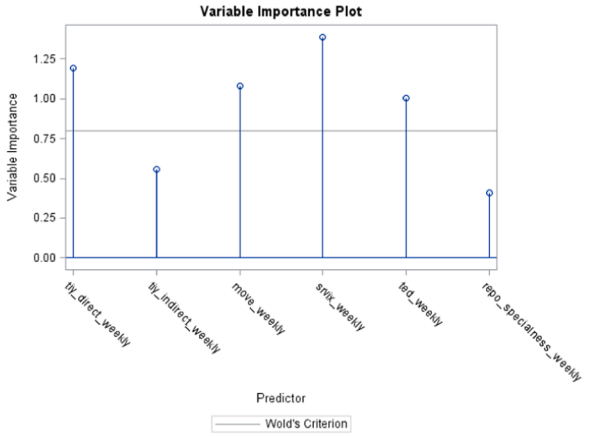


Fig. 5. Variable importance plot.
We consider Treasury inconvenience yield (direct holdings), Treasury inconvenience yield (indirect holdings), MOVE, SRVIX, TED, and repo specialness. The dots in the plot represent the scores for the first two factors. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies.

Table 12
Bai and Perron's (1998, 2003) test for structural breaks.

Regime	Sample span	TIPS discounts (basis points, bps)			
		Mean	SD	Min	Max
1	Aug. 1, 2012–Sep. 4, 2015	7.91	2.47	3.10	16.84
2	Sep. 7, 2015–Jan. 15, 2021	8.37	2.15	4.75	16.75
3	Jan. 18, 2021–Dec. 30, 2022	7.41	3.37	3.10	16.84

This table summarizes the results of Bain and Perron's (1998, 2003) test for structural breaks. SD denotes the standard deviation. IQR is interquartile range. We aggregate the daily data from August 1, 2012, to December 30, 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies. TIPS discounts are in basis points.

Table 13
Subsample analysis.

	Regime 1	Regime 2	Regime 3
Marking-to-market concerns	0.3745***	0.3353***	0.4185***
	[6.51]	[4.56]	[4.47]
Intermediation frictions	0.1253***	0.0809	−0.0251
	[3.09]	[1.63]	[−0.59]
Inflation swap rate	0.3659***	0.5830***	−0.0236
	[7.02]	[9.36]	[−0.28]
Return on the S&P500 index	−0.0091	−0.0417**	0.0258
	[−0.65]	[−2.13]	[1.23]
VIX	−0.0589	−0.1539**	0.0008
	[−1.41]	[−2.39]	[0.02]
BAA10Y-CMT10Y	0.0721	−0.0394	0.0261**
	[1.24]	[−0.44]	[2.34]
Year-month fixed effects	Yes	Yes	Yes
N	480	254	94
F statistic	60.15***	60.63***	123.04***
R^2	0.9575	0.9599	0.9821

This table presents the results of the multivariable regression analysis for each subsample listed in Table 12. We standardize all the variables, making it possible to directly compare the magnitude of the estimated coefficients. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies. N denotes the number of observations. Heteroscedasticity-consistent test statistics are shown in square brackets. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

inflation, the unique feature of the principal-adjusting mechanism indicates that one needs to hold a TIPS issue until maturity to get fully compensated against inflation. This restriction raises marking-to-market concerns when the economic situation is uncertain. Hence, although the TIPS–Treasury bond mispricing averages 3.18 % per year, it does not disappear. The study's results suggest that when arbitrage is costly, asset prices could deviate from fundamental values for prolonged periods (Gemmill and Thomas, 2002).

Table 14
Inflation, Federal fund rates, and TIPS discounts.

	(1)	(2)	(3)
Marking-to-market concerns	0.5198***	0.4954***	0.4604***
	[9.22]	[7.78]	[7.18]
Intermediation frictions	0.0604	0.0012	−0.0061
	[1.41]	[0.03]	[−0.14]
Marking-to-market concerns × Inflation swap rate	−0.0131		−0.0101
	[−1.08]		[−0.72]
Intermediation frictions	0.1366***		0.0673**
× Inflation swap rate	[4.91]		[2.12]
Marking-to-market concerns × effective Federal fund rate		0.0032	0.0113
		[0.24]	[0.71]
Intermediation frictions		0.1429***	0.1229***
× effective Federal fund rate		[7.44]	[5.78]
Return on the S&P500 index	0.0029	−0.0561	−0.0639
	[0.21]	[−1.37]	[−1.55]
VIX	−0.1263***	0.0009	−0.0001
	[−3.04]	[0.07]	[−0.00]
BAA10Y-CMT10Y	0.3519***	0.3331***	0.3586***
	[6.14]	[6.17]	[6.38]
Year-month fixed effects	Yes	Yes	Yes
N	480	480	480
F statistic	57.12***	62.08***	61.71***
R^2	0.9575	0.9608	0.9613

This table presents the results of the multivariable regression analysis where we allow both marking-to-market concerns and intermediation frictions to vary directly with the inflation swap rate and the effective Federal fund rate. We standardize all the variables, making it possible to directly compare the magnitude of the estimated coefficients. We aggregate the daily data from January 4, 2010 to December 30, 2022 by week and compute weekly averages to minimize market microstructural noises and house-specific idiosyncrasies. N denotes the number of observations. Heteroscedasticity-consistent test statistics are shown in square brackets. ***, **, and * represent statistical significance at the 1 %, 5 %, and 10 % levels, respectively.

5. Concluding remarks

TIPS are usually undervalued compared to cash flow-matched Treasury bonds. We use J.P. Morgan's proprietary asset swap data to explore the factors associated with TIPS discounts. We also quantify the relative importance of potential drivers regarding the magnitude of the price gaps between TIPS and Treasury bonds. Our study shows that uncertainty-related variables are key to understanding the underpricing of TIPS issues compared to their hypothetical counterparts. We conclude that when strategic concerns; for example, marking-to-market concerns, overwhelm fundamental analysis, asset prices could deviate from fundamental values for prolonged periods.

In real terms, Treasury coupon securities are risky. Treasury bills are risky to long-term investors. Inflation-indexed securities fill this gap, as they offer several benefits. First, TIPS protect investors against inflation. TIPS facilitate investors' portfolio choices and address market participants' concerns regarding inflation. Second, TIPS enable issuers to save on the inflation risk premium inherent in nominal issuances. However, these benefits are yet to be realized, as TIPS discounts remain in a prolonged horizon. Persistent TIPS discounts offset the advantage of bearing no inflation risk premium as issuers incur additional borrowing costs, even if liquidity significantly improves as the market matures. Our results suggest that intermediary frictions and strategic concerns could better moderate the TIPS–Treasury bond puzzle.

CRediT authorship contribution statement

Jungkyu Ahn: Conceptualization, Data curation, Methodology, Supervision, Visualization, Writing – original draft, Writing – review & editing. **Yongkil Ahn:** Conceptualization, Formal analysis, Methodology, Software, Validation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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